

Exploring Foot Pressure Sensing for Avatar Manipulation and for Input Modality in VR

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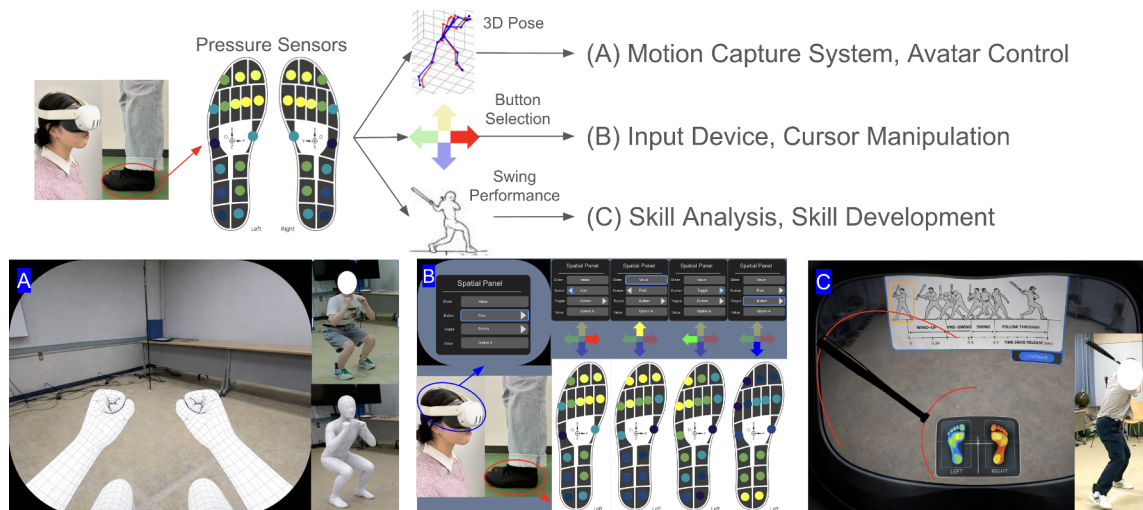


Figure 1: Foot Pressure Input and Three Application in VR. A; avatar manipulation by pose estimation using foot pressure sensor, B; crosshair-input, C; baseball swinging practice application

ABSTRACT

Avatar manipulation is a fundamental component of virtual reality (VR) systems, as it enables immersive user experiences and supports natural and expressive interaction. Foot pressure sensing is as a promising alternative input for lightweight and unobtrusive. Because foot-ground interaction naturally occurs during common VR activities such as standing, walking, and weight shifting, pressure-based signals can be captured with minimal instrumentation while encoding rich information about whole-body dynamics. These properties make foot pressure sensing particularly well suited for VR scenarios that prioritize seamless integration, user comfort, and embodied interaction.

In this paper, we extend dataset to more variety of motion and group them into four categories, hand manipulation, squat and jump, arm swing and locomotion. Then, we trained transformer-based network model. We analyzed accuracy by each category. We observe that hand pose estimation is consistently less accurate than lower-body estimation, with errors becoming more pronounced during rapid movements. This degradation is likely due to

the combination of high motion velocity, limited sensing coverage for the upper body, and the absence of direct force-based signals for the hands.

Finally, we propose three application of VR with foot pressure sensor that span input, embodiment, and skill training. Specifically, we employ pressure signals as a hands-free input modality for both discrete and continuous control, enable full-body avatar manipulation through pressure-based pose estimation without extensive upper-body instrumentation, and support sports-oriented interaction by analyzing lower-body weight transfer during baseball swing practice. Together, these applications illustrate how foot pressure sensing can facilitate expressive, unobtrusive, and context-aware avatar manipulation across diverse VR scenarios.

Index Terms: Foot pressure, pose estimation, dataset, virtual reality

1 INTRODUCTION

Avatar manipulation is a core component of virtual reality (VR) systems, as it enables immersive user experiences and supports natural and expressive interaction within virtual environments. In VR, an avatar can be manipulated either by directly mapping a user's physical motion to the virtual body or by interpreting explicit user commands that trigger discrete actions. Achieving accurate and responsive avatar manipulation is critical, as tighter synchronization between a user's movements and the avatar's behavior significantly enhances the sense of embodiment and body ownership in VR experiences[6, 5].

Existing VR systems, for example Meta Quest series or HTC Vive, primarily rely on handheld controllers, vision-based

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tracking, or inertial measurement units (IMUs) to manipulate avatars. Controller-based interaction allows users to issue commands through buttons and triggers. Vision-based and IMU-based approaches, in contrast, infer avatar motion and interaction intent from a user's body movements or predefined gesture patterns[3, 4], enabling more natural interaction. More recent advances further explore bio-sensing modalities, such as electromyography (EMG), to capture muscle activity for fine-grained avatar control[1, 2].

In contrast to upper-body- or vision-centric approaches, foot pressure sensing provides a promising alternative for lightweight and unobtrusive avatar manipulation. Foot-ground interaction naturally occurs during common VR activities such as standing, walking, and weight shifting, and pressure signals can be captured with minimal additional instrumentation. These characteristics make foot pressure sensing well suited for VR scenarios where seamless integration and user comfort are critical. Prior studies demonstrate that pressure distributions encode rich information about gait dynamics and whole-body movement[7], suggesting strong potential for enhancing avatar control.

In this paper, we enhance dataset to apply more variety of motions and investigate accuracy between different types of motion. Moreover, we suggest more application of foot pressure sensors to VR.

2 METHOD AND IMPLEMENTATION

In Fig2, we show overview of our implementation of pose estimation by foot pressure sensor. First we collect foot pressure data, and then we reduce the number of sensor channels. Then we input those values to neural network system and attain prediction.

2.1 Data Collection

We recruited 6 volunteers (M=4, F=2; body mass: 54.5 ± 7.1 kg; body height: 165.3 ± 5.9 cm; age: 24.3 ± 4.4 years old). The participants are university student and postdoctoral fellow, any of them is not a professional sports player nor does not engaged in a physically demanding occupation. All participants were physically healthy and without chronic chronic at the time of the study. For the capturing, participants were instructed to move freely within a lab room. The research is conducted with an approval by the Human Subject for Science and Engineering Research Ethics Review Committee and the President of Institute of Science Tokyo.

We collected four motion types, hand manipulation, squat and jump, arm swing and locomotion. Each category aims to different type of activities.

The hand manipulation category includes motions where participant does not change foot position and only moves their hands. The squat and jump category includes motions where participants move vertically and bend both knees together. The arm swing category includes motions where participant twists body and moves a arm or arms quickly while holding an object, usually dominant hand or both hands. The locomotion category consists of motions that participants move within a lab.

The sensor used to collect foot pressures is XSENSOR¹, a 235-ch foot pressure sensor. This streams recorded values to a computer via Bluetooth. The 3D pose data is captured by 4 GoPro HERO13 Blacks² and analyzed by Move AI³, a markerless motion capture system. We capture video with and synchronize video. Then, we process by Move AI with M2 model. The motion data is synchronized with pressure sensors and down sampled by 60 Hz.

2.2 Network

As shown in Fig. 3, we employ a multi-head causal Transformer encoder as our baseline model. The model takes insole pressure data as input, written as $F \in \mathbb{R}^{c \times f}$, where c is the number of pressure sensors, and f denotes the number of samples within a 1/3 second window at 60 Hz. The Transformer encodes temporal dependencies through self-attention over F and estimates the pose representation $P \in \mathbb{R}^{3 \times f \times j}$, where $j = 17$ represents the number of body joints.

The model is trained by minimizing the following combined loss:

$$\mathcal{L}_n = \mathcal{L}_n^{\text{MSE}}(y_n, \hat{y}_n) + \lambda \mathcal{L}_n^{\text{MSE}}(y_n - y_{n-1}, \hat{y}_n - \hat{y}_{n-1}), \quad (1)$$

where $\mathcal{L}^{\text{MSE}}$ denotes the mean squared error loss. The first term measures the position error between the predicted and ground-truth joint positions, while the second term penalizes the difference in joint velocities to reduce motion jitter. In our experiments, the weight λ is set to 10 to balance pose accuracy and temporal smoothness.

The model is trained end-to-end using PyTorch frameworks. We use a batch size of 32 and update the weights using the AdamW optimizer with a learning rate of 1.0×10^{-4} . We train our model for 70 epochs (less than 20 hours) on an NVIDIA RTX 4090 GPU.

2.3 Training Result

In Tab1, we show accuracy of each joint by different motion category. Accuracy is evaluated using the Mean Per Joint Position Error (MPJPE), a widely used metric that measures the average Euclidean distance between predicted and ground-truth joint positions. This analysis allows us to identify how different types of motion influence the reliability of joint estimation.

The results show that hand manipulation tasks yield the lowest MPJPE values across most joints. This outcome can be attributed to the relatively low dynamic range of these movements. In contrast, arm swing exhibit the highest MPJPE and noticeable jitter at wrist. This pattern aligns with the inherently abrupt and high-velocity nature of swinging motions, where rapid acceleration and deceleration introduce significant frame-to-frame variation. Such rapid transitions make it difficult for the model to maintain temporal consistency, thus amplifying prediction noise and increasing positional error.

Overall, these findings highlight that pose estimation accuracy is closely related to the kinematic characteristics of each motion category. Motions involving smooth and localized movement, such as hand manipulation, favor lower estimation error, while highly dynamic or multi-joint coordinated actions, such as swinging and jumping, tend to challenge model stability. These observations suggest that improving temporal smoothing or incorporating motion-specific priors may help reduce joint-level jitter in future models.

In addition, we evaluate inference time for predicting one posture from 20 frames, 0.4 seconds, of data. The inference time with Apple Silicon M2 CPU is 16.1 milliseconds on average.

3 APPLICATION PROPOSAL

Foot pressure sensing provides several advantages that make it particularly well suited for avatar manipulation in VR systems. In this section, we explain foot pressure sensor itself and our methods's features and advantages. Then, we propose potential application.

¹<https://www.xsensor.com/>

²<https://gopro.com/en/us/shop/cameras/learn/hero13black/CHDHX-131-master.html>

³<https://www.move.ai/>

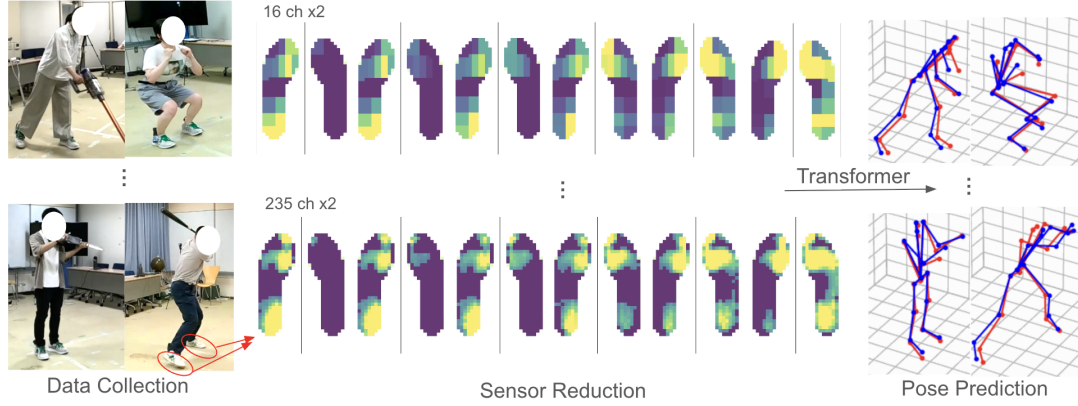


Figure 2: Components for Our Implementation; we first conduct data collection. Then, we apply filters and train models.

Table 1: Result of Training Using All Sensor by Joints

Motion Category	All (average of all joint)		Wrist		Ankle	
	MPJPE (cm)	Jitter (cm^2/s^3)	MPJPE (cm)	Jitter (cm^2/s^3)	MPJPE (cm)	Jitter (cm^2/s^3)
Hand Manipulation	6.5	1.24×10^6	12.8	2.61×10^6	5.6	1.38×10^6
Squat and Jump	11.3	2.69×10^6	16.2	3.78×10^6	14.1	4.36×10^6
Arm Swing	13.0	2.38×10^6	24.4	5.21×10^6	14.4	3.23×10^6
Locomotion	13.8	2.20×10^6	21.1	3.27×10^6	17.7	3.54×10^6
ALL	13.2	2.21×10^6	22.4	3.91×10^6	15.7	3.23×10^6

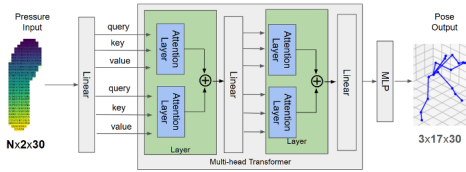


Figure 3: Our network structure

3.1 Foot Pressure sensor feature in VR

To begin with, pressure sensors can be seamlessly embedded into footwear or insoles, resulting in an input modality that is lightweight, easily worn, and largely imperceptible to the user. Because the sensing hardware does not introduce visible or bulky components, foot pressure-based interaction preserves immersion and avoids disrupting the user's embodied experience.

From an interaction perspective, foot pressure sensing naturally aligns with foot-oriented avatar actions, such as locomotion, balance adjustment, and directional control. In foot oriented motion like locomotion, lower-body input can feel intuitive and engaging. Particularly in applications where foot movement plays a central role, such as role-playing games or exploratory VR environments, foot input can enhance immersion.

Moreover, foot pressure sensors provide access to kinetic information that is difficult to obtain from commonly used inertial sensing alone. While IMUs capture kinematic signals such as acceleration and angular velocity, pressure distributions reflect forces and load transfer between the user and the ground. This kinetic information enables more reliable inference of postural and contextual states, including sitting, standing, riding in an elevator, or ascending stairs, which are challenging to distinguish using IMU data alone.

In addition, Foot pressure-based interaction also reduces so-

cial awkwardness of using VR in public. Unlike mid-air hand gestures or pinch-based commands, which can be conspicuous to surrounding observers, foot pressure input remains largely invisible. This subtlety allows users to interact comfortably without self-consciousness, addressing an often-overlooked barrier to the adoption of immersive VR systems outside private settings.

3.2 Application Proposal

Based on the capabilities of the proposed foot pressure sensing system, we explore three representative VR applications that demonstrate its versatility for avatar manipulation and interaction.

First, we propose foot pressure sensing as crosshair-based cursor interface, an alternative input modality for discrete and continuous control. In this application, pressure signals are mapped to a crosshair-based cursor interface, where the magnitude and temporal variation of applied force determine the speed at which the selection target changes. This design enables hands-free interaction while allowing fine-grained control through natural weight shifting, making it suitable for scenarios where the user's hands are occupied or where reduced upper-body motion is desirable.

Second, we apply foot pressure sensing to full-body avatar manipulation through pose estimation. As explained in Section 3.1, pose estimation by foot pressure sensors enables continuous control of an avatar's posture and movement without relying on extensive upper-body instrumentation. This approach supports more natural and immersive avatar embodiment, as lower-body signals play a central role in locomotion, balance, and posture regulation, which are critical components of believable full-body avatar control in VR.

Finally, we demonstrate a sports-oriented application in the form of baseball swing training. Effective baseball swinging requires precise coordination between lower-body weight transfer, upper-body rotation, and bat movement. Foot pressure sensing provides direct insight into how a player shifts weight across different regions of the foot during a swing, enabling detailed analysis and

feedback on technique. Combined with full-body motion estimation, this application highlights the potential of pressure-based sensing for skill training and performance analysis in VR environments.

4 DISCUSSION

4.1 Discussion

In this paper, we introduced a dataset that includes both rapid body motions and motions involving object interactions, and we evaluated model performance across multiple activity categories. Our analysis reveals distinct accuracy trends across body parts and motion types. In particular, we observe that hand pose estimation is consistently less accurate than lower-body estimation, with errors becoming more pronounced during rapid movements. In contrast, rapid lower-body movements—such as those observed in squatting and jumping actions—are estimated with comparatively higher accuracy.

This degradation is likely due to the combination of high motion velocity, limited sensing coverage for the upper body, and the absence of direct force-based signals for the hands. The robustness observed in lower-body estimation highlights the effectiveness of pressure-based sensing for activities where force distribution and weight transfer play a dominant role, reinforcing its suitability for lower-body-centric avatar manipulation.

Despite these promising results, our work has several limitations. First, all experiments were conducted on a single, flat terrain, and we did not evaluate performance across varying ground conditions such as slopes, uneven surfaces, or compliant floors. Terrain variation may significantly influence pressure distributions and, consequently, estimation accuracy. Second, the network architecture and loss formulation employed in this study are intentionally simple. Incorporating joint-specific error weighting or more advanced loss designs may lead to improved performance, especially for visually salient joints such as the hands. These limitations point to several promising directions for future work. Additionally, extending the model to support seated postures, such as avatar manipulation while sitting on a chair, represents an important step toward broader applicability in everyday VR scenarios, including office work and social VR experiences.

5 CONCLUSION

This paper investigated foot pressure sensing as a lightweight and unobtrusive modality for avatar manipulation in virtual reality. By leveraging naturally occurring foot-ground interactions, pressure-based signals provide rich information about whole-body dynamics while requiring minimal instrumentation, making them well suited for VR scenarios that prioritize comfort, immersion, and embodied interaction.

To evaluate the capabilities and limitations of this approach, we extended an existing dataset to include a wider range of motions and organized them into four activity categories: hand manipulation, squat and jump, arm swing, and locomotion. Using a transformer-based network, we analyzed estimation accuracy across these categories and observed a consistent performance gap between upper- and lower-body motion. In particular, hand pose estimation exhibited higher error, especially during rapid movements, highlighting the challenges of inferring fine-grained upper-body dynamics from lower-body pressure signals alone.

Beyond quantitative evaluation, we demonstrated the applicability of foot pressure sensing through three representative VR use cases spanning hands-free input, full-body avatar manipulation, and sports skill training. These applications illustrate how pressure-based sensing can support expressive interaction, robust embodiment, and context-aware feedback without relying on extensive upper-body or visual instrumentation. Together, our findings suggest that foot pressure sensing is a viable and complementary

modality for avatar manipulation in VR and motivate future work on richer sensing configurations, improved modeling of upper-body motion, and broader interaction scenarios.

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